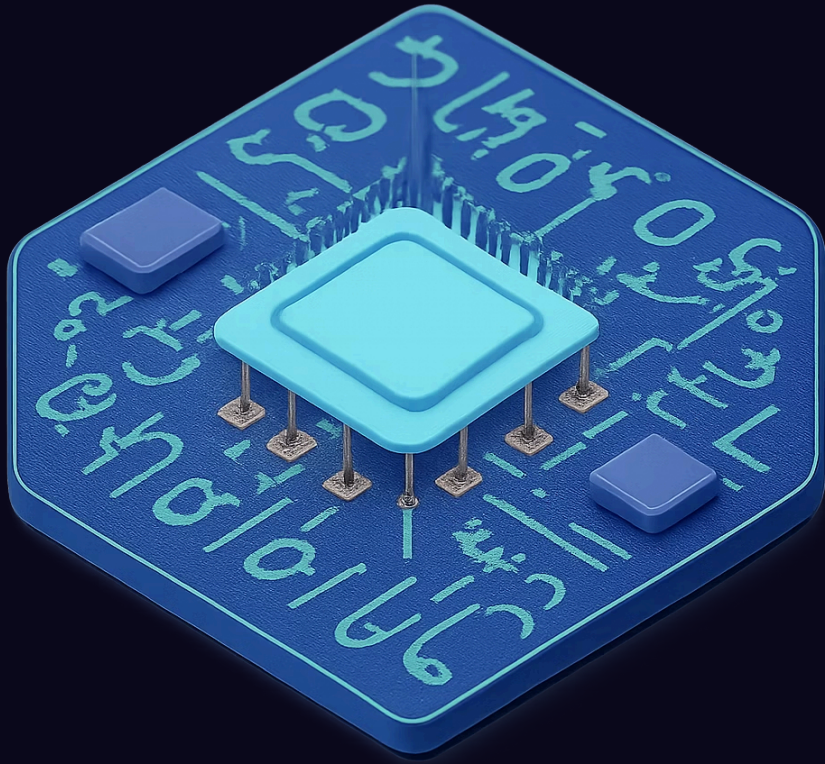




Design and Simulation of an 8-Bit Signed Arithmetic Unit

1st-Year B.Tech DDCA project

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Academic year 2026

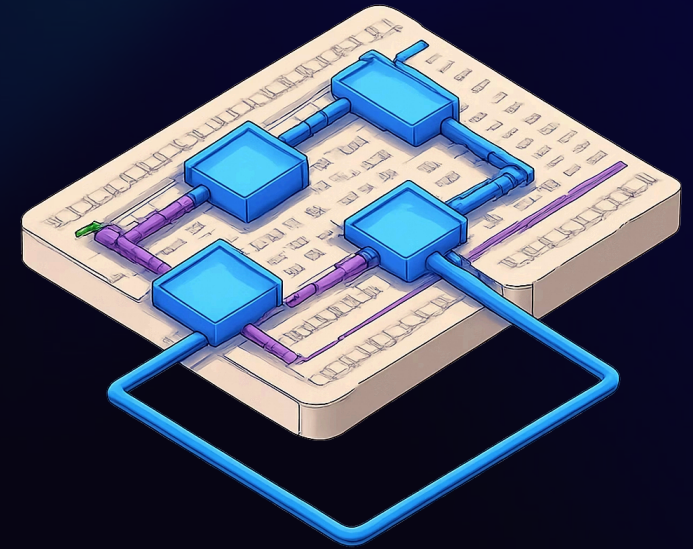
Problem Statement & Abstract

What Problem Does This Project Solve?

- Computers need to do math calculations very quickly
- They must handle both **positive and negative** numbers
- The system must be small and use less power

Why Do We Need This Project?

- Basic calculators **cannot** handle negative numbers easily
- Microprocessors are too complex for beginners to understand
- We need a simple, educational system to learn digital design
- To design a **fast and efficient** arithmetic system from scratch



Objectives

1

Design

Create an 8-bit signed arithmetic unit using logic gates and adders

2

Implement

Build the circuit design using digital design tools

3

Simulate

Test the system using computer simulation software

4

Verify

Check that the system works correctly for all test cases

5

Document

Write clear explanations so anyone can understand the project

Literature Survey & Existing Systems



What Makes Our Project Different?

- Handles **both positive and negative** numbers
- Simpler than microprocessors — easy to understand
- Fast and efficient operation
- **Perfect for learning** digital design basics

Gap We Are Filling

- No simple educational system existed to learn from
- We bridge the gap between basic adders and complex CPUs

Overview & What is 8-Bit ?



Arithmetic Unit

A small system that does basic math — addition and subtraction. Found inside **every computer and calculator**.



Signed Numbers

Positive: 0 to 127

Negative: -1 to -128

Handled using Two's Complement method.

01
10

8-Bit Means

Works with **8 binary digits** (0s and 1s). Handles numbers from **-128 to +127**.



Project Aim

Design, simulate, and verify an arithmetic unit that adds and subtracts 8-bit signed numbers.

System Architecture & Block Diagram



Main Components

Input Module

Receives two 8-bit numbers and operation selection

Control Unit

Decides whether to add or subtract

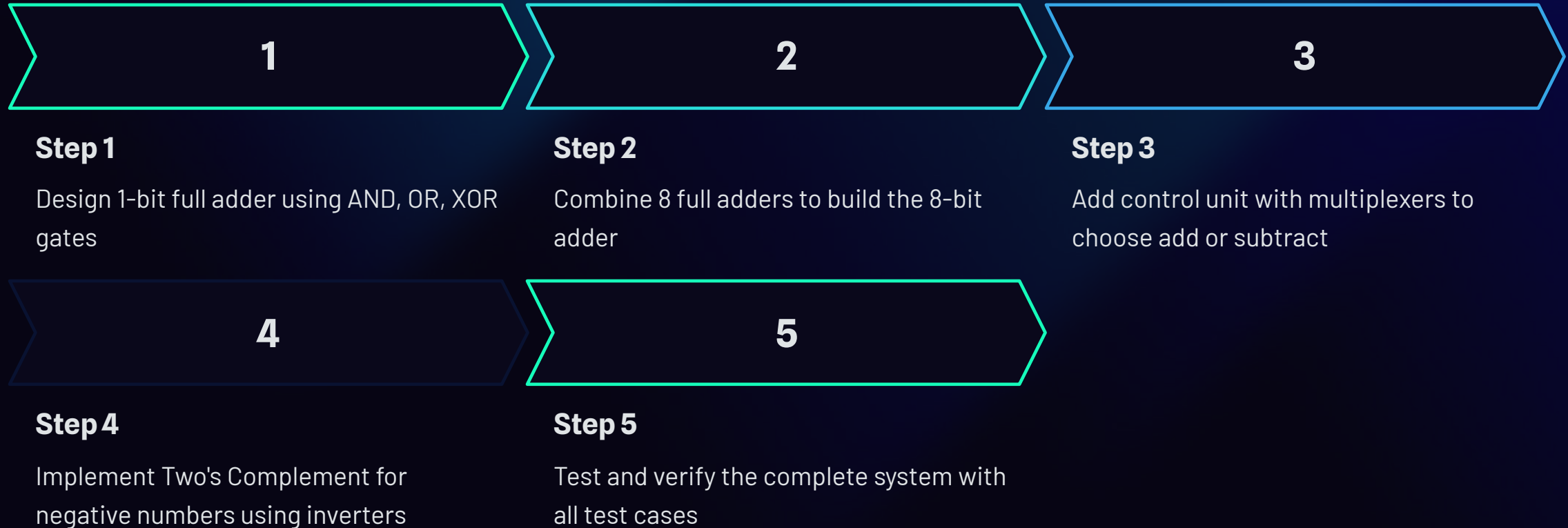
ALU

Uses adders & logic gates to process numbers

Output Module

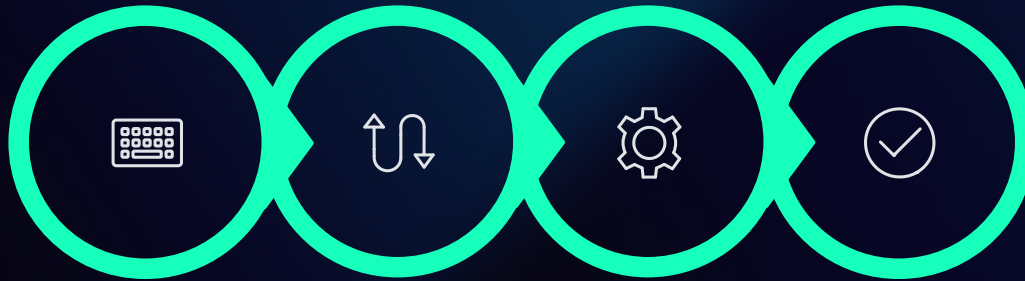
Displays the final 8-bit result

Methodology & Implementation



📌 **Key Concept:** Subtraction is done as $A - B = A + (-B)$. The second number is inverted and 1 is added to get its negative. Example: $+5 = 00000101$, $-5 = 11111011$

How It Works — Step by Step



Give Numbers

Choose
Operation

System
Processes

Get Answer

Addition Example Subtraction Example

The system handles both positive and negative numbers **automatically** using the Two's Complement method.

$$5 + 3 = 8 \checkmark$$

Input: 00000101 + 00000011

Output: 00001000

$$5 + (-3) = 2 \checkmark$$

B is inverted using Two's Complement, then added to A

Simulation & Testing Results

Our 8-bit signed arithmetic unit underwent rigorous simulation and testing using various positive and negative number combinations for both addition and subtraction. All test cases passed successfully, demonstrating the unit's accuracy and robustness.

Test Case	Operation	Input (Binary)	Expected Output	Actual Output	Status
1	$5 + 3 = 8$	00000101 + 00000011	00001000	00001000	✓
2	$-5 + 3 = -2$	11111011 + 00000011	11111110	11111110	✓
3	$10 - 7 = 3$	00001010 - 00000111	00000011	00000011	✓
4	$15 + (-8) = 7$	00001111 + 11111000	00000111	00000111	✓
5	$3 - 10 = -7$	00000011 - 00001010	11111001	11111001	✓

These results confirm that the arithmetic unit correctly handles both positive and negative integer operations within the 8-bit signed range using the Two's Complement representation.

Conclusion & Future Scope

What We Achieved

- Designed a working 8-bit signed arithmetic unit
- Successfully handled positive and negative numbers
- Verified correct results using simulation

What We Learned

- How computers do math at the basic level
- How logic gates combine to build larger systems
- How Two's Complement represents negative numbers

Future Scope

- Extend to 16-bit or 32-bit operations
- Add multiplication and division
- Implement on actual hardware (FPGA)

Thank You!

